



US 20250285802A1

(19) **United States**(12) **Patent Application Publication**
TADA et al.(10) **Pub. No.: US 2025/0285802 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **TRANSFORMER MODULE**(52) **U.S. Cl.**CPC **H01F 27/292** (2013.01); **H05K 1/181**
(2013.01)(71) Applicant: **KABUSHIKI KAISHA TOYOTA**
JIDOSHOKKI, Kariya-shi (JP)(72) Inventors: **Yoshifumi TADA**, Kariya-shi (JP);
Katsuya USAMI, Kariya-shi (JP);
Atsushi OSHIMA, Kariya-shi (JP)

(57)

ABSTRACT(73) Assignee: **KABUSHIKI KAISHA TOYOTA**
JIDOSHOKKI, Kariya-shi (JP)(21) Appl. No.: **19/063,478**(22) Filed: **Feb. 26, 2025**(30) **Foreign Application Priority Data**

Mar. 7, 2024 (JP) 2024-035230

Publication Classification(51) **Int. Cl.****H01F 27/29** (2006.01)**H05K 1/18** (2006.01)

A transformer module includes multiple lead terminals, each of which includes a bent portion bent along a substrate and a joint portion extending along the substrate. Lands that are arranged in parallel on the substrate include an enlarged land and standard lands. Joint portions of the lead terminals, which are adjacent to each other in a parallel arrangement direction of the lead terminals and have the same electric potential, are soldered to the enlarged land via a solder layer. Each joint portion is soldered to one of the standard lands via the corresponding solder layer. The enlarged land is formed by connecting, via a connection land and in the parallel arrangement direction, the standard lands to which the lead terminals, adjacent to each other in the parallel arrangement direction and having the same electric potential, are intended to be soldered.

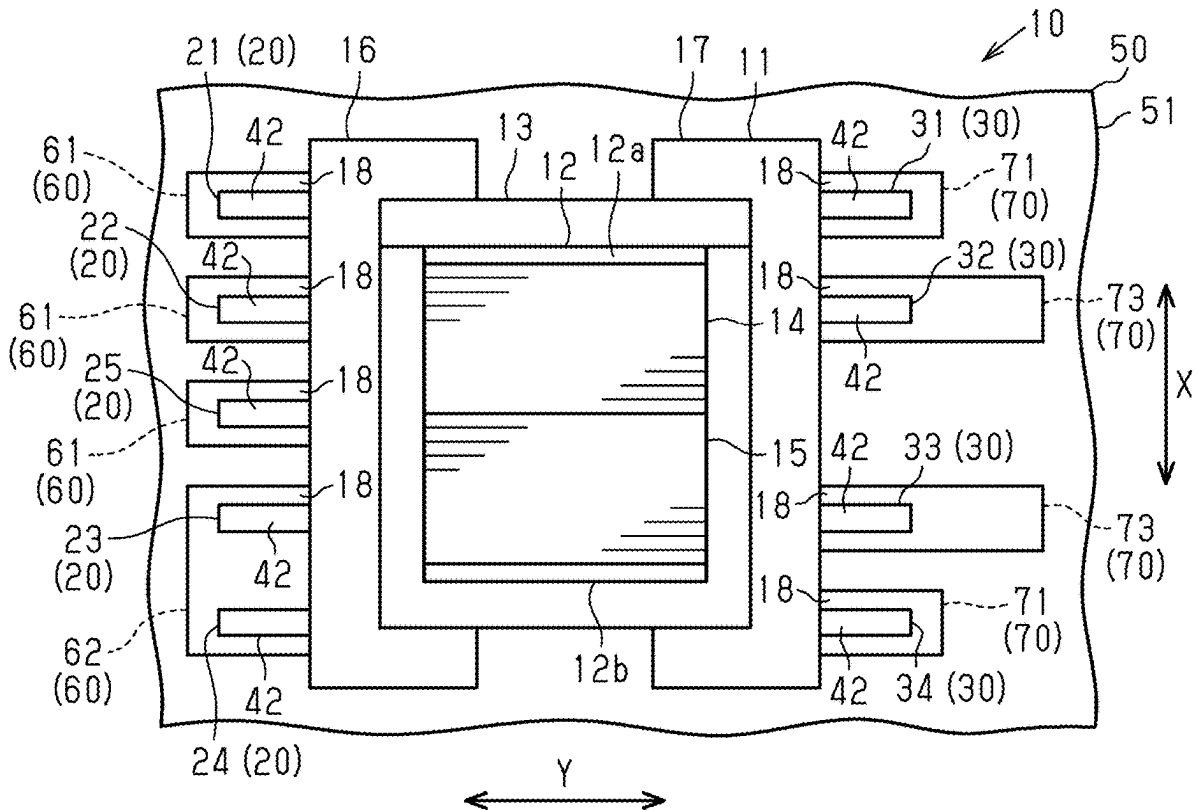


Fig.1

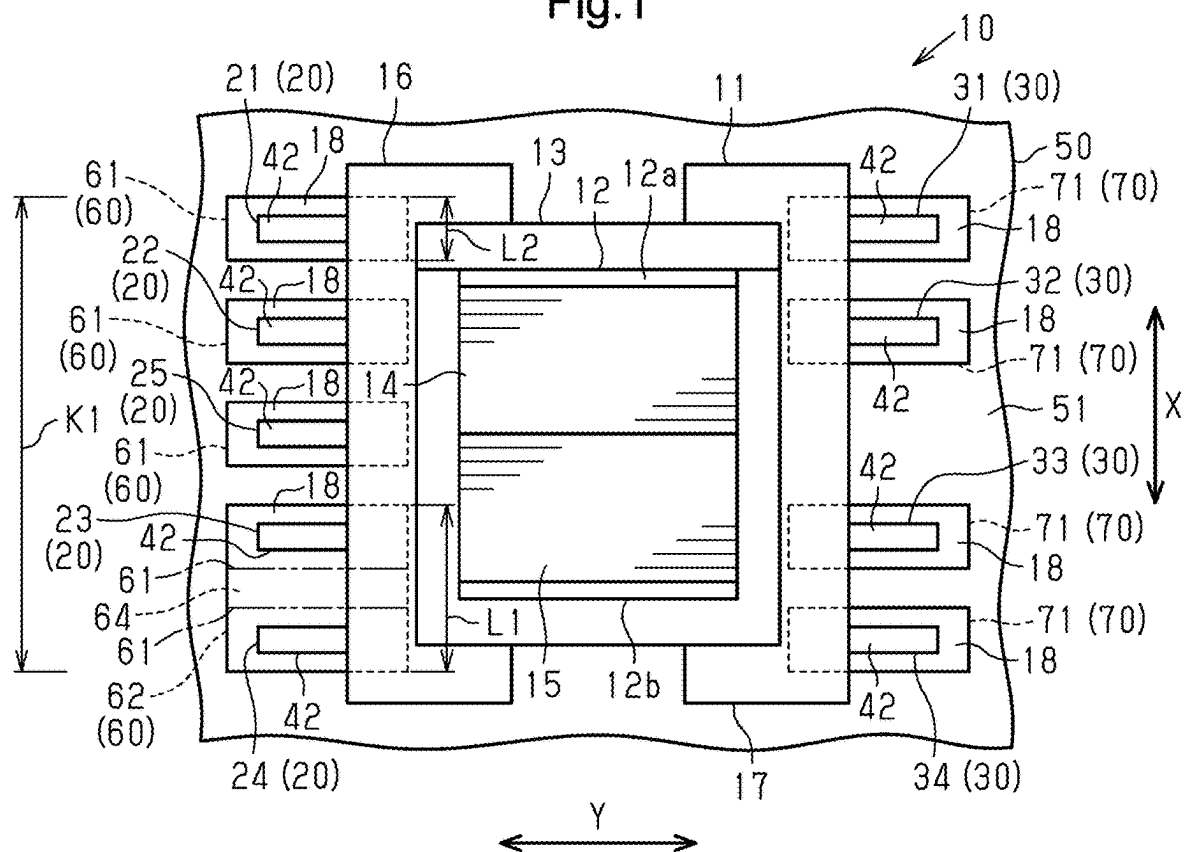


Fig.2

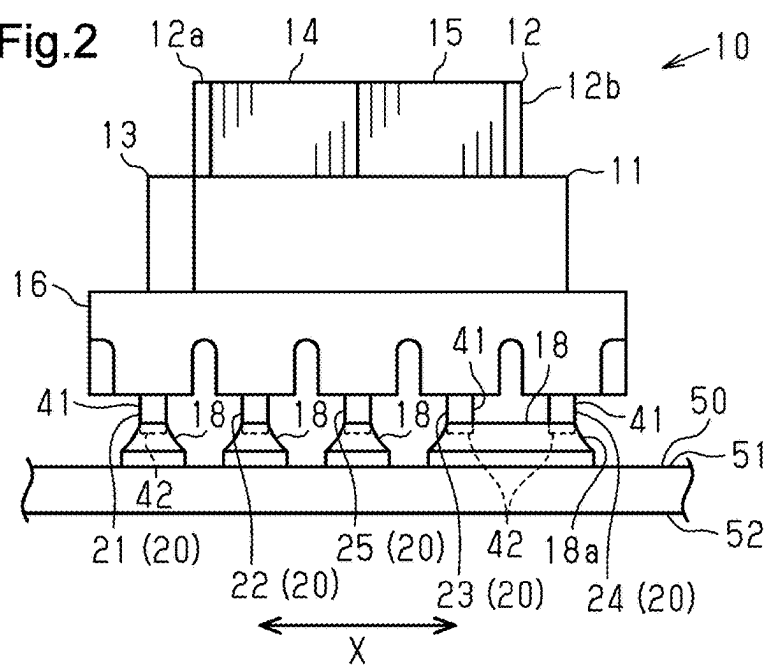


Fig.3

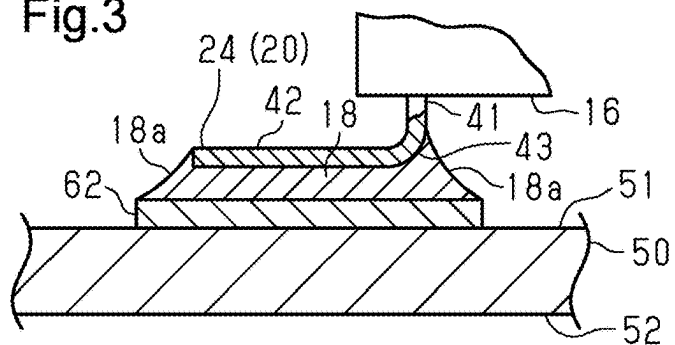


Fig.4

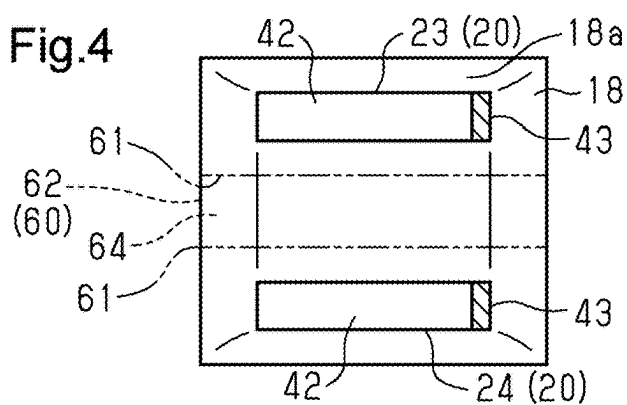


Fig.5

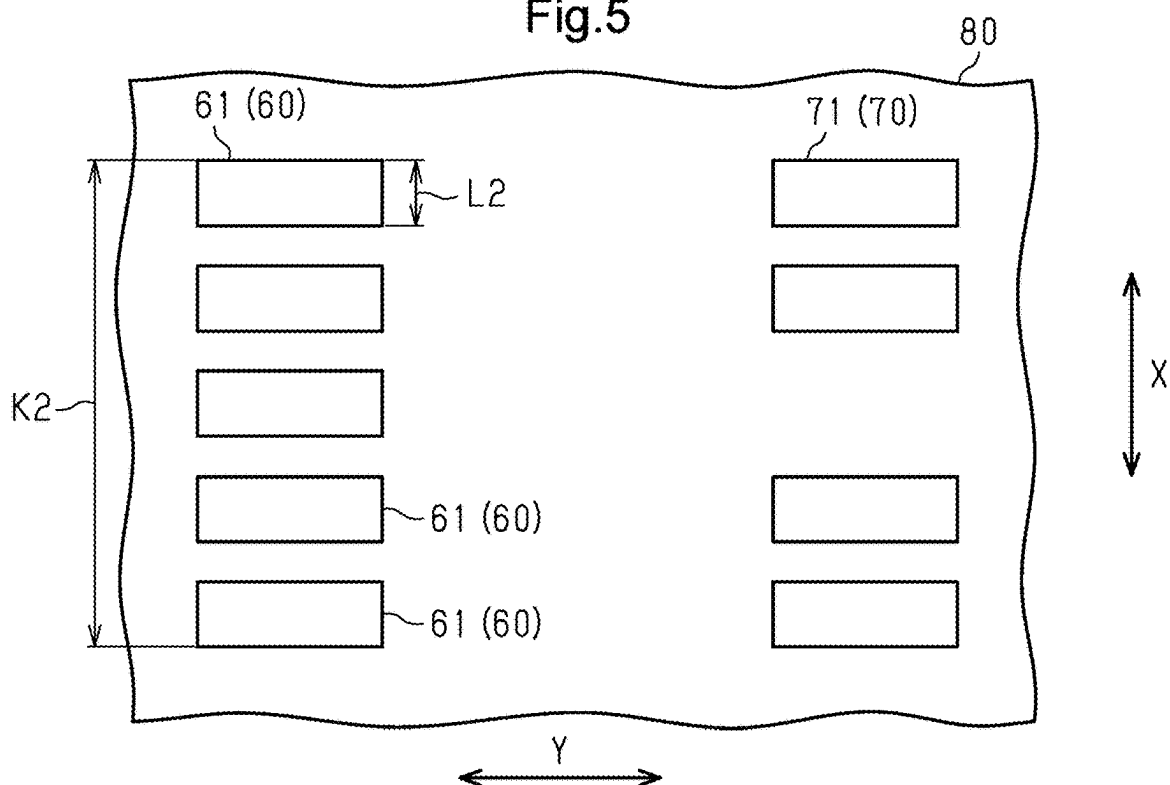


Fig.6

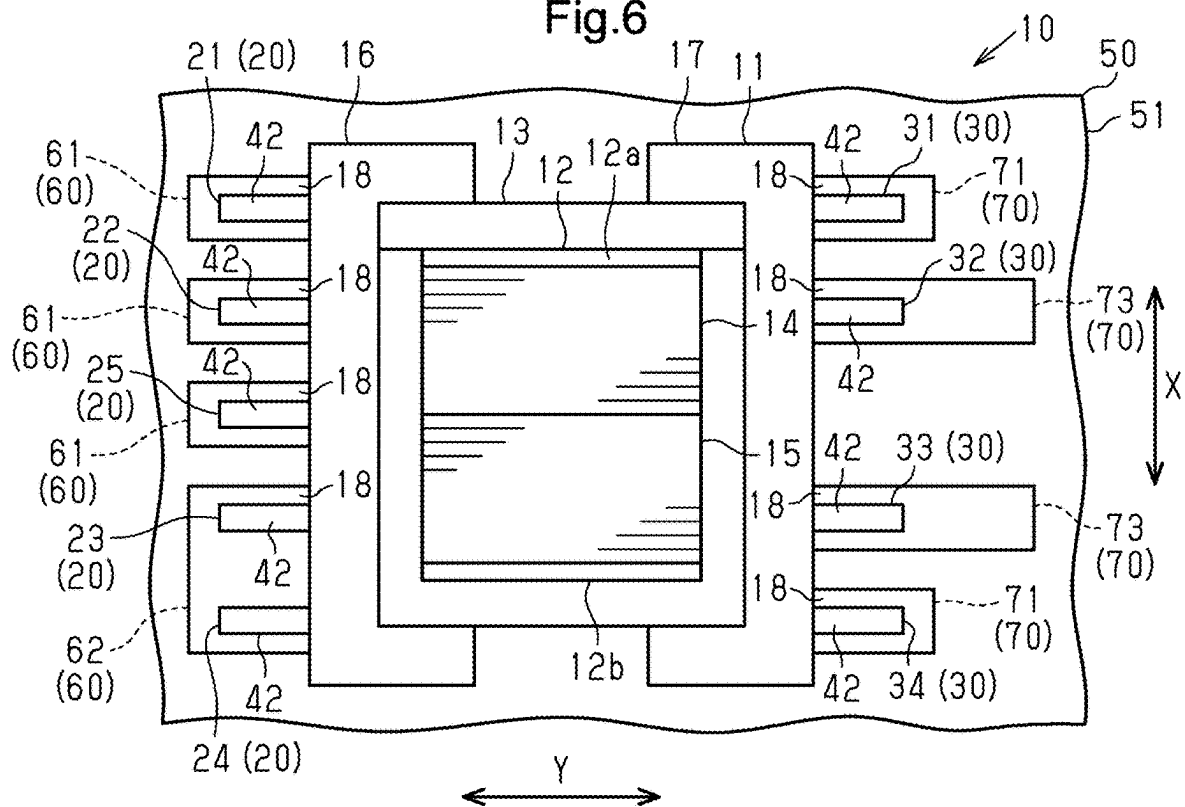
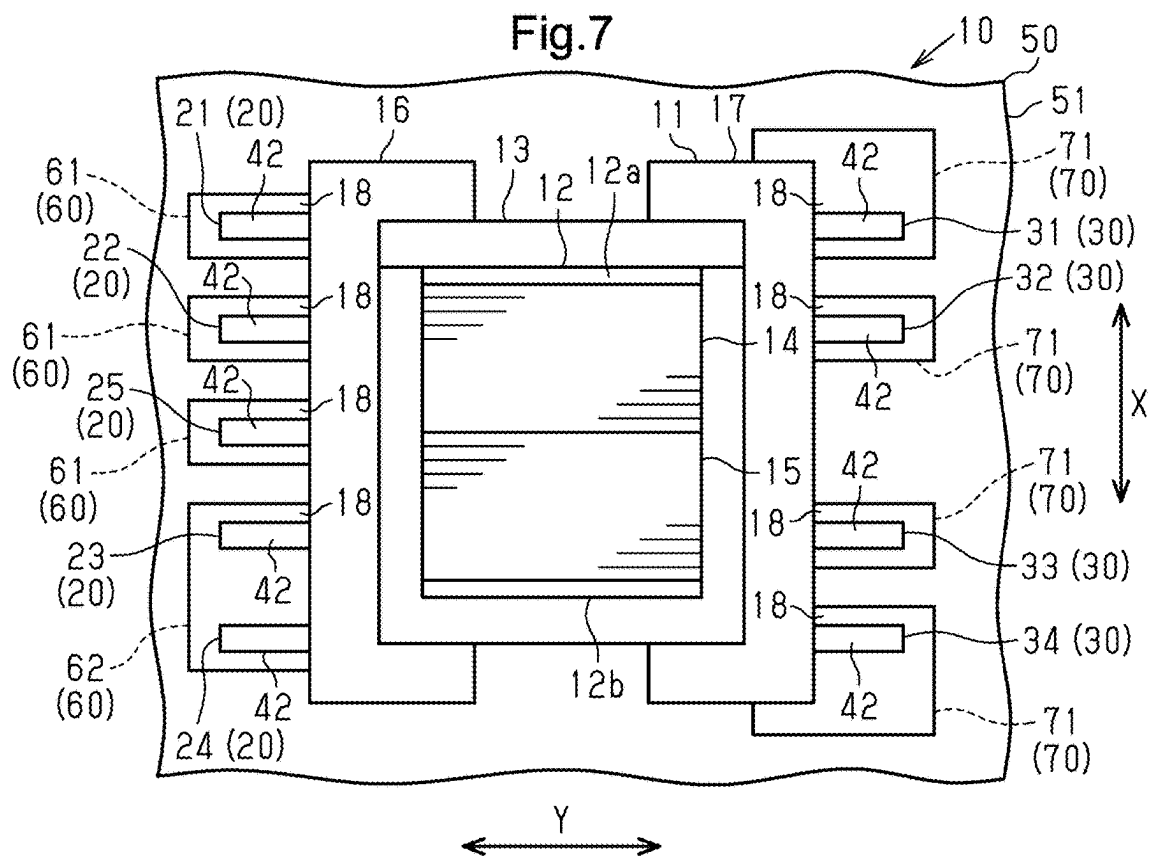


Fig.7



TRANSFORMER MODULE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-035230, filed on Mar. 7, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

[0002] The present disclosure relates to a transformer module.

2. Description of Related Art

[0003] The transformer module includes a substrate and a transformer. The transformer comprises a winding and multiple lead terminals electrically connected to the winding. The lead terminals are arranged in parallel. Each of the lead terminals is bent from its root, and a portion including its tip is soldered to a land of the substrate via a solder layer.

[0004] In a transformer module, the transformer is a relatively large component. Consequently, when the transformer module is subjected to external vibrations or similar forces, the transformer is prone to vibration as well. The bent portions of the lead terminals are locations where stress is likely to occur when the transformer vibrates. Therefore, it is preferable to reinforce the bent portions of the lead terminals in the transformer module. One method of reinforcement involves joining a portion of the solder layer to the bent portion. In order to join a portion of the solder layer to the bent portion, it is necessary to increase the amount of solder paste applied to the land. As a method of increasing the amount of solder paste applied to the land, for example, the technique disclosed in Japanese Laid-Open Patent Publication No. H7-131139 is known. In this publication, a wiring board includes a land for soldering terminals and an auxiliary land provided continuously with the land. Since the solder paste applied to the auxiliary land moves toward the land, the amount of the solder paste supplied to the land is increased.

[0005] However, as disclosed in the above publication, adding an auxiliary land to the substrate results in an increase in the overall size of the substrate.

SUMMARY

[0006] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0007] In one general aspect, a transformer module includes a transformer and a substrate. The transformer includes multiple lead terminals electrically connected to corresponding windings, and a terminal mount on which the multiple lead terminals are arranged in parallel. Multiple lands to which the lead terminals are respectively soldered are arranged in parallel on the substrate. Each of the lead terminals includes a bent portion bent along the substrate and a joint portion extending along the substrate. Each joint portion is soldered to the corresponding land via a solder

layer. The lands include an enlarged land and multiple standard lands. The joint portions of ones of the lead terminals are soldered to the enlarged land via the corresponding solder layer. The ones of the lead terminals are adjacent to each other in a parallel arrangement direction in which the lead terminals are arranged in parallel. The ones of the lead terminals have a same electric potential. Each joint portion is soldered to one of the standard lands via the corresponding solder layer. The enlarged land is formed by connecting, via a connection land and in the parallel arrangement direction, the standard lands to which the lead terminals, adjacent to each other in the parallel arrangement direction and having the same electric potential, are intended to be soldered.

[0008] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a plan view showing a transformer module according to an embodiment.

[0010] FIG. 2 is a side view of the transformer module shown in FIG. 1.

[0011] FIG. 3 is an enlarged view showing a lead terminal and a solder layer in the enlarged land shown in FIG. 1.

[0012] FIG. 4 is a plan view showing the lead terminals and the solder layer in the enlarged land shown in FIG. 3.

[0013] FIG. 5 is a plan view showing a substrate of a reference example.

[0014] FIG. 6 is a plan view showing a transformer module according to a first modification.

[0015] FIG. 7 is a plan view showing a transformer module according to a second modification.

[0016] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0017] This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

[0018] Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

[0019] In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

[0020] A transformer module 10 according to an embodiment will now be described with reference to FIGS. 1 to 5.

Transformer Module

[0021] As shown in FIGS. 1 and 2, the transformer module 10 includes a transformer 11 and a substrate 50, to which the transformer 11 is soldered.

[0022] The transformer 11 includes a bobbin 12, a magnetic core 13 combined with an outer peripheral portion of the bobbin 12, and a primary winding 14 and a secondary winding 15 wound around the bobbin 12. The bobbin 12 includes a flange portion 12a at one end in the axial direction and a flange portion 12b at the other end in the axial direction. The primary winding 14 and the secondary winding 15 are wound around the bobbin 12 such that the one flange portion 12a, the primary winding 14, the secondary winding 15, and the other flange portion 12b are arranged in that order in the axial direction. The number of turns is different between the primary winding 14 and the secondary winding 15. The magnetic core 13 may be an EI-type core or an EE-type core. A primary-side terminal mount 16 and a secondary-side terminal mount 17 are formed below the bobbin 12. Each of the primary-side terminal mount 16 and the secondary-side terminal mount 17 has the shape of a block, and their longitudinal directions agree with each other.

[0023] The primary-side terminal mount 16 is provided with five primary-side lead terminals 20. The five primary-side lead terminals 20 include a first lead terminal 21, a second lead terminal 22, a dummy lead terminal 25, a third lead terminal 23, and a fourth lead terminal 24, which are arranged in that order from one end to the other end in the longitudinal direction of the primary-side terminal mount 16. The first lead terminal 21 to the fourth lead terminal 24 and the dummy lead terminal 25 are arranged in parallel on the primary-side terminal mount 16 at equal intervals in the longitudinal direction of the primary-side terminal mount 16. Therefore, the direction in which the five primary-side lead terminals 20 are arranged in parallel agrees with the longitudinal direction of the primary-side terminal mount 16.

[0024] Any two of the primary-side lead terminals 20 that are adjacent to each other in the longitudinal direction of the primary-side terminal mount 16 are separated from each other by a distance greater than or equal to the insulation distance. The first lead terminal 21 to the fourth lead terminal 24 are electrically connected to the primary winding 14. The first lead terminal 21 and the second lead terminal 22, which are adjacent to each other in the longitudinal direction of the primary-side terminal mount 16, have the same electric potential, and the third lead terminal 23 and the fourth lead terminal 24, which are adjacent to each other in the longitudinal direction of the primary-side terminal mount 16, have the same electric potential.

[0025] Accordingly, the transformer 11 included in the transformer module 10 includes the first lead terminal 21 to the fourth lead terminal 24, which are electrically connected to the primary winding 14, and the first lead terminal 21 to the fourth lead terminal 24 are arranged in parallel on the primary-side terminal mount 16.

[0026] The secondary-side terminal mount 17 is provided with four secondary-side lead terminals 30. The four secondary-side lead terminals 30 include a first lead terminal 31, a second lead terminal 32, a third lead terminal 33, and a fourth lead terminal 34, which are arranged in that order from one end to the other end in the longitudinal direction of the secondary-side terminal mount 17. The first lead

terminal 31 to the fourth lead terminal 34 are arranged in parallel in the longitudinal direction of the secondary-side terminal mount 17. The parallel arrangement direction of the four secondary-side lead terminals 30 agrees with the longitudinal direction of the secondary-side terminal mount 17 and also agrees with the parallel arrangement direction of the five primary-side lead terminals 20.

[0027] The first lead terminal 31 to the fourth lead terminal 34 are electrically connected to the secondary winding 15. The first lead terminal 31 and the second lead terminal 32, which are adjacent to each other in the longitudinal direction of the secondary-side terminal mount 17, have different electric potentials, and the third lead terminal 33 and the fourth lead terminal 34, which are adjacent to each other in the longitudinal direction of the secondary-side terminal mount 17, have different electric potentials. The second lead terminal 32 and the third lead terminal 33, which are adjacent to each other in the longitudinal direction of the secondary-side terminal mount 17, have different electric potentials.

[0028] Accordingly, the transformer 11 included in the transformer module 10 includes the first lead terminal 31 to the fourth lead terminal 34, which are electrically connected to the secondary winding 15, and the first lead terminal 31 to the fourth lead terminal 34 are arranged in parallel on the secondary-side terminal mount 17.

[0029] In the transformer 11, for example, when AC input power is input to the primary winding 14 through the first lead terminal 21 and the fourth lead terminal 24 on the primary side, an output voltage is generated at the first lead terminal 31 and the third lead terminal 33 on the secondary side. In the transformer 11, when AC input power is input to the primary winding 14 through the second lead terminal 22 and the third lead terminal 23 on the primary side, an output voltage is generated at the second lead terminal 32 and the fourth lead terminal 34 on the secondary side. An input voltage is converted into an output voltage in accordance with the ratio of the number of turns between the primary winding 14 and the secondary winding 15.

[0030] Each of the primary-side lead terminals 20 and the secondary-side lead terminals 30 is formed by bending a thin plate into an L-shape. The primary-side lead terminals 20 and the secondary-side lead terminals 30 have the same shape. Therefore, in the following description, the same component names and reference numerals are assigned to the components included in both the primary-side lead terminals 20 and the secondary-side lead terminals 30.

[0031] As shown in FIGS. 3 and 4, each of the primary-side lead terminals 20 and the secondary-side lead terminals 30 integrally includes a rectangular plate-shaped extending portion 41, a rectangular plate-shaped joint portion 42, and a bent portion 43. The extending portion 41 extends from the primary-side terminal mount 16 or the secondary-side terminal mount 17 on which the terminal 20, 30 is disposed. The joint portion 42 extends so as to intersect with the extending portion 41. The bent portion 43, connects the extending portion 41 and the joint portion 42 to each other.

Substrate

[0032] As shown in FIGS. 1 and 2, the substrate 50 includes a first main surface 51 and a second main surface 52, which are located on opposite sides in the thickness direction. The substrate 50 includes lands provided on the first main surface 51. The lands include multiple primary-

side lands **60** and multiple secondary-side lands **70**. In plan view in which the transformer module **10** is viewed in the thickness direction, the primary-side lands **60** are disposed at positions corresponding to the primary-side terminal mount **16** of the transformer **11**, and the secondary-side lands **70** are disposed at positions corresponding to the secondary-side terminal mount **17** of the transformer **11**. In plan view in which the transformer module **10** is viewed in the thickness direction, the primary-side lands **60** and the secondary-side lands **70** are disposed with the transformer **11** interposed therebetween.

[0033] The primary-side lead terminals **20** are soldered to the primary-side lands **60** via solder layers **18**. Accordingly, the primary-side lands **60**, to which the primary-side lead terminals **20** are respectively soldered, are arranged in parallel on the substrate **50**. The secondary-side lead terminals **30** are soldered to the secondary-side lands **70** via the corresponding solder layers **18**. Accordingly, the secondary-side lands **70**, to which the secondary-side lead terminals **30** are respectively soldered, are arranged in parallel on the substrate **50**.

[0034] Each of the primary-side lands **60** has a rectangular shape. The primary-side lands **60** are arranged such that the short sides of all the primary-side lands **60** extend in the same direction. The direction in which the short sides of the primary-side lands **60** extend is defined as a first direction **X** of the substrate **50**. The direction in which the long sides of the primary-side lands **60** extends is defined as a second direction **Y** of the substrate **50**. The primary-side lands **60** are arranged in parallel in the first direction **X** of the substrate **50**. Accordingly, the parallel arrangement direction of the primary-side lands **60** agrees with the first direction **X** of the substrate **50**. Any two of the primary-side lands **60** that are adjacent to each other in the first direction **X** of the substrate **50** are separated from each other in the first direction **X** by a distance greater than or equal to the insulation distance.

[0035] The primary-side lands **60** include standard lands **61** to each of which one joint portion **42** is soldered via the solder layer **18**. The primary-side lands **60** include an enlarged land **62** to which two joint portions **42** are soldered. On the substrate **50**, three standard lands **61** and one enlarged land **62** are arranged in parallel from one end to the other end in the first direction **X** of the substrate **50**.

[0036] Each standard land **61** has a surface area sufficient to allow the application of an amount of solder paste necessary for soldering a single joint portion **42** to the primary-side land **60** via a solder layer **18**. This amount of solder paste is sufficient to form the solder layer **18** so that it joins to the entire back surface and side surfaces of the joint portion **42** while securing the joint portion **42** to the primary-side land **60**.

[0037] The standard lands **61** and the enlarged land **62** have the same dimension in the second direction **Y**. The standard lands **61** and the enlarged land **62** have different dimensions in the first direction **X**. A dimension **L1** of the enlarged land **62** in the first direction **X** is greater than a dimension **L2** of the standard land **61** in the first direction **X**. The dimension **L1** of the enlarged land **62** is sufficiently greater than twice the dimension **L2** of the standard land **61**.

[0038] A substrate **80** of a reference example shown in FIG. 5 will now be described.

[0039] The substrate **80** of the reference example includes five standard lands **61** as the primary-side lands **60** and four

standard lands **71** as the secondary-side lands **70**. The dimensions in the first direction **X** of the standard lands **61**, **71** in the reference example are the same as the dimension **L2** of the standard lands **61**, **71** in the embodiment. The dimensions in the second direction **Y** of the standard lands **61**, **71** in the reference example are the same as the dimension in the second direction **Y** of the standard lands **61**, **71** in the embodiment. The distance between any two of the standard lands **61**, **71** adjacent to each other in the first direction **X** is also the same as the distance between any two of the standard lands **61**, **71** adjacent to each other in the first direction **X** in the embodiment. On the substrate **80** of the reference example, two standard lands **61** are provided in place of the enlarged land **62**.

[0040] As shown in FIGS. 1 and 4, the enlarged land **62** of the embodiment is formed by connecting two standard lands **61** that are located at one end in the first direction **X** and adjacent to each other in the first direction **X** on the substrate **80** of the reference example. Specifically, as indicated by the long-dash double-short-dash lines in FIG. 1, the enlarged land **62** is formed by connecting, with a connection land **64** and in the parallel arrangement direction, two standard lands **61** to which the third lead terminal **23** and the fourth lead terminal **24**, having the same potential and being adjacent to each other in the parallel arrangement direction, are respectively intended to be soldered. These two standard lands **61** are connected to each other over the entire length of the standard lands **61** via the connection land **64**. That is, the connection land **64** connects the standard lands **61** adjacent to each other in the parallel arrangement direction over the entire standard lands **61** in the direction orthogonal to the parallel arrangement direction. Accordingly, the dimension of the enlarged land **62** in the second direction **Y** of the substrate **50** is the same as the dimension of the standard land **61** in the second direction **Y**, while the dimension of the enlarged land **62** in the first direction **X** of the substrate **50** is sufficiently greater than twice the dimension of the standard land **61** in the first direction **X**. The surface area of the enlarged land **62** is the sum of the surface areas of the two standard lands **61** and the surface area of the connection land **64**. Therefore, the surface area of the enlarged land **62** is sufficiently greater than twice the surface area of each standard land **61**.

[0041] The surface area of the enlarged land **62** and the surface area of the standard land **61** refer to the surface area of the enlarged land **62** and the surface area of the standard land **61**, respectively, in plan view of the first main surface **51** of the substrate **50** as viewed in the thickness direction.

[0042] As shown in FIG. 1, on the substrate **50** of the embodiment, the distance in the first direction **X** between the outer edges of the primary-side lands **60** located at the opposite ends in the first direction **X** is referred to as a standard distance **K1**. As shown in FIG. 5, on the substrate **80** of the reference example, the distance in the first direction **X** between the outer edges of the standard lands **61** located at the opposite ends in the first direction **X** is referred to as a reference distance **K2**. The standard distance **K1** is equal to the reference distance **K2**. Accordingly, the substrate **50** of the embodiment includes the enlarged land **62** without increasing, in the first direction **X**, the surface area of the region in which the primary-side lands **60** are disposed, as compared to the substrate **80** of the reference example. Furthermore, the substrate **50** of the embodiment does not increase the surface area of the region in which the

primary-side lands **60** are disposed in the second direction Y, as compared to the substrate **80** of the reference example.

[0043] As shown in FIG. 1, each of the secondary-side lands **70** has a rectangular shape. The secondary-side lands **70** are arranged such that the short sides of all the secondary-side lands **70** extend in the first direction X. The secondary-side lands **70** are arranged in parallel in the first direction X of the substrate **50**. Any two of the secondary-side lands **70** that are adjacent to each other in the first direction X of the substrate **50** are separated from each other in the first direction X by a distance greater than or equal to the insulation distance.

[0044] Each of the four secondary-side lands **70** is a standard land **71** to which one secondary-side lead terminal **30** is soldered. The primary-side lands **60** and the secondary-side lands **70** are disposed at positions opposite to each other in the second direction Y of the substrate **50**. Specifically, at one end in the first direction X of the substrate **50**, two standard lands **61** and two standard lands **71** are disposed on the opposite sides from each other in the second direction Y. At the other end in the first direction X of the substrate **50**, the enlarged land **62** and two standard lands **71** are disposed on the opposite sides from each other in the second direction Y. No secondary-side land **70** is disposed at a position aligned in the second direction Y with the standard land **61** at the center of the five primary-side lands **60** arranged in parallel in the first direction X.

Transformer Module

[0045] The transformer module **10**, which includes the transformer **11** having the above-described configuration and the substrate **50**, will now be described.

[0046] As shown in FIG. 1, on the primary side of the transformer **11**, each of the first lead terminals **21** and the second lead terminals **22** is soldered to the corresponding standard land **61** of the primary-side lands **60** via a solder layer **18**. On the primary side of the transformer **11**, the joint portions **42** of two of the primary-side lead terminals **20** are soldered to the enlarged land **62** via a solder layer **18**. These two primary-side lead terminals **20** are adjacent to each other in the parallel arrangement direction, in which the multiple primary-side lead terminals **20** are arranged in parallel, and have the same electric potential. Therefore, the transformer module **10** includes the enlarged land **62** on the substrate **50**. The enlarged land **62** is soldered, via a solder layer **18**, to the joint portions **42** of the third lead terminal **23** and the fourth lead terminal **24**. The third lead terminal **23** and the fourth lead terminal **24** are adjacent to each other in the parallel arrangement direction, in which the primary-side lead terminals **20** are arranged in parallel, and have the same electric potential. The dummy lead terminal **25** is joined via a solder layer **18** to the standard land **61** located at the center in the parallel arrangement direction of the substrate **50**. On the secondary side of the transformer **11**, the first lead terminal **31** to the fourth lead terminal **34** are joined to the standard lands **71** of the secondary-side lands **70** via the solder layers **18**.

[0047] As shown in FIG. 2, the extending portion **41** of each of the primary-side lead terminals **20** soldered to the primary-side lands **60** extends from the lower surface of the primary-side terminal mount **16** toward the substrate **50**. Although not illustrated, the extending portion **41** of each of the secondary-side lead terminals **30** soldered to the sec-

ondary-side lands **70** extends from the lower surface of the secondary-side terminal mount **17** toward the substrate **50**.

[0048] As shown in FIG. 3, each of the primary-side lead terminals **20** includes a bent portion **43** that is bent along the substrate **50**, and a joint portion **42** that extends along the substrate **50** and is soldered to the corresponding primary-side land **60** via a solder layer **18**. Although not illustrated, each of the secondary-side lead terminals **30** includes a bent portion **43** that is bent along the substrate **50**, and a joint portion **42** that extends along the substrate **50** and is soldered to the corresponding secondary-side land **70** via a solder layer **18**.

[0049] As shown in FIGS. 3 and 4, the solder layer **18** is joined to the joint portions **42** over the entire length of the joint portions **42**, and is also joined to the distal end faces of the joint portions **42** and the outer surfaces of the bent portions **43**. The solder layer **18** is joined to the opposite side surfaces in the transverse direction of each joint portion **42**. Thus, the solder layer **18** is joined to the joint portions **42** so as to surround the joint portions **42**.

[0050] The solder layer **18** that solders the third lead terminal **23** and the fourth lead terminal **24** to the enlarged land **62** includes solder fillet **18a** extending downward while being inclined so as to spread from the upper edge of the joint portion **42** toward the outer edge of the standard land **61**. That is, the joint portions **42** are joined to the upper portion of the solder layer **18**, which is formed so as to rise on the enlarged land **62**. The solder fillet **18a** extends downward while being inclined so as to spread from the upper ends of the bent portions **43** toward the outer edge of the enlarged land **62**. Therefore, the solder layer **18** is joined to the bent portions **43** over the entire height of the bent portions **43**. That is, the solder layer **18** covers the entire height of the bent portions **43**.

Operation of the Embodiment

[0051] As a specific example of a method of soldering the transformer **11** to the substrate **50**, a method of soldering the third lead terminal **23** and the fourth lead terminal **24** to the enlarged land **62** will be described together with operation of the transformer module **10** according to the present embodiment.

[0052] First, a solder paste is applied to each of the primary-side lands **60** and the secondary-side lands **70**. Next, the joint portions **42** of the primary-side lead terminals **20** are placed on the solder paste applied to the primary-side lands **60**, and the joint portions **42** of the secondary-side lead terminals **30** are placed on the solder paste applied to the secondary-side lands **70**.

[0053] As shown in FIG. 1, since the enlarged land **62** is formed by connecting two standard lands **61** via the connection land **64**, the surface area of the enlarged land **62** is sufficiently greater than twice the surface area of each standard land **61**. Accordingly, the amount of solder paste that can be applied to the enlarged land **62** is increased. As the amount of the solder paste applied to the enlarged land **62** increases, the solder paste is more likely to flow toward the joint portions **42**, and thus the solder paste is more likely to flow toward the bent portions **43**. As a result of the solder paste rising along the joint portions **42** and the bent portions **43**, the solder layer **18** is joined to the bent portions **43** over the entire height.

Advantages of the Embodiment

[0054] The above-described embodiment has the following advantages.

[0055] (1) On the substrate 50, the enlarged land 62 is formed by connecting two standard lands 61 arranged in the parallel arrangement direction via the connection land 64. Therefore, the amount of solder paste that can be applied to the enlarged land 62 is increased as compared with a case in which the solder paste is applied only to one standard land 61. The third lead terminal 23 and the fourth lead terminal 24, which have the same electric potential, are soldered to the enlarged land 62 via a solder layer 18. The solder layer 18 is thus joined to the bent portion 43 of each of the third lead terminal 23 and the fourth lead terminal 24. Consequently, the solder layer 18 reinforces the bent portions 43, in which stress is likely to occur due to vibration or the like.

[0056] The enlarged land 62 is formed by connecting, via the connection land 64, two standard lands 61, to which the third lead terminal 23 and the fourth lead terminal 24 of the same electric potential are intended to be soldered. Since the primary-side lands 60 to which the primary-side lead terminals 20 having the same electric potential are soldered do not need to be insulated from each other, the adjacent primary-side lands 60 can be connected to each other via the connection land 64 disposed therebetween. In other words, the enlarged land 62 enlarges the surface area of a land by effectively using the region between the adjacent standard lands 61. Therefore, even though the enlarged land 62 is formed, the arrangement region of the primary-side lands 60 on the substrate 50 is not enlarged in the parallel arrangement direction of the primary-side lands 60. Accordingly, the bent portions 43 are reinforced without increasing the size of the substrate 50.

[0057] (2) The solder layer 18 is joined to the third lead terminal 23 and the fourth lead terminal 24, which are soldered to the enlarged land 62, over the entire height of the bent portions 43 via the solder fillet 18a. This reinforces the bent portions 43 in a favorable manner.

[0058] (3) The transformer 11 includes the dummy lead terminal 25. The dummy lead terminal 25 is provided in the transformer 11 as a primary-side lead terminal 20. The number of the primary-side lead terminals 20 including the dummy lead terminal 25 is five, and the number of the secondary-side lead terminals 30 is four. Thus, since the primary-side lead terminals 20 and the secondary-side lead terminals 30 can be clearly distinguished from each other, it is possible to prevent the primary-side lead terminals 20 from being soldered to the secondary-side lands 70.

[0059] (4) The enlarged land 62 is formed by connecting two standard lands 61 via the connection land 64 over the entire length of the standard lands 61. The enlarged land 62 thus has a rectangular shape. Accordingly, for example, the surface area of the enlarged land 62 is increased as compared with a case in which an H-shaped enlarged land is formed by connecting only parts of the two standard lands 61 in the longitudinal direction via the connection land 64. Consequently, the amount of solder paste that can be applied to the enlarged land 62 is sufficiently increased. In addition, the fluidity of the solder paste applied to the enlarged land 62 is increased as compared with a case in which only parts of the two standard lands 61 in the longitudinal direction are connected via the connection land 64. The solder paste thus readily flows toward the joint portions 42 and the bent portions 43. Accordingly, the bent portions 43 are reinforced

by the solder layer 18 formed of the solder paste applied to the enlarged land 62 in a favorable manner.

Modifications

[0060] The above-described embodiment may be modified as follows. The above-described embodiment and the following modifications can be combined as long as the combined modifications remain technically consistent with each other.

[0061] As shown in FIG. 6, as part of the secondary-side lands 70, elongated lands 73 having a longer dimension in the second direction Y than the standard lands 71 may be provided on the substrate 50. Such elongated lands may be provided on the substrate 50 as part of the primary-side lands 60. In such a configuration, the amount of solder paste that can be applied to each elongated land 73 is increased as compared with the amount of solder paste applied to each standard land 71. Accordingly, also in each elongated land 73, the solder layer 18 is joined to the bent portion 43 of the primary-side lead terminal 20 or the secondary-side lead terminal 30. Consequently, the solder layer 18 reinforces the bent portions 43, in which stress is likely to occur due to vibration or the like.

[0062] As shown in FIG. 7, among the secondary-side standard lands 71 arranged in parallel in the first direction X, each of the standard lands 71 arranged at the opposite ends in the parallel arrangement direction may be increased in dimension in a direction away from the adjacent standard land 71. Among the standard lands 61 of the primary-side land 60, the standard land 61 disposed at one end in the parallel arrangement direction may also be increased in dimension in a direction away from the adjacent standard land 61.

[0063] Accordingly, even a lead terminal other than the third lead terminal 23 and the fourth lead terminal 24, which have the same electric potential, can be soldered to the corresponding standard land 71 with an increased amount of solder paste. Since the dimension of the standard land 71 arranged at least at one end in the parallel arrangement direction is increased in a direction away from the adjacent standard land 71, the standard land 71 arranged at least at one end does not short-circuit with the adjacent standard land 71. Therefore, the bent portion 43 of the secondary-side lead terminal 30 joined to the standard land 71 is readily reinforced.

[0064] In the primary-side lands 60, the first lead terminal 21 and the second lead terminal 22, which are arranged in the parallel arrangement direction and have the same electric potential, may be soldered to the enlarged land 62.

[0065] In the above-described embodiment, one of the five primary-side lead terminals 20 is used as the dummy lead terminal 25. However, that one lead terminal may be used as a lead terminal that is energized in the same manner as the other four lead terminals of the primary-side lead terminals 20.

[0066] In the transformer module 10, the number of the primary-side lead terminals 20 and the number of the secondary-side lead terminals 30 arranged in the parallel arrangement direction may be other than those in the embodiment. The number of the primary-side lead terminals 20 or the secondary-side lead terminals 30 that are adjacent to each other in the parallel arrangement direction and have the same electric potential may be three or more. According to the number of the primary-side lead terminals 20 or the

secondary-side lead terminals **30** of the same electric potential, the enlarged land **62** may be formed by connecting the same number of the standard lands **61** as the number of the primary-side lead terminals **20** or the secondary-side lead terminals **30** of the same electric potential via a connection land **64**.

[0067] Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

What is claimed is:

1. A transformer module, comprising:

a transformer including multiple lead terminals electrically connected to corresponding windings, and a terminal mount on which the multiple lead terminals are arranged in parallel; and

a substrate on which multiple lands to which the lead terminals are respectively soldered are arranged in parallel, wherein

each of the lead terminals includes:

a bent portion bent along the substrate; and

a joint portion extending along the substrate,

each joint portion is soldered to the corresponding land via a solder layer,

the lands include an enlarged land and multiple standard lands,

the joint portions of ones of the lead terminals are soldered to the enlarged land via the corresponding solder layer, the ones of the lead terminals being adjacent to each other in a parallel arrangement direction in which the lead terminals are arranged in parallel, and the ones of the lead terminals having a same electric potential,

each joint portion is soldered to one of the standard lands via the corresponding solder layer, and

the enlarged land is formed by connecting, via a connection land and in the parallel arrangement direction, the standard lands to which the lead terminals, adjacent to each other in the parallel arrangement direction and having the same electric potential, are intended to be soldered.

2. The transformer module according to claim 1, wherein the standard lands are arranged in parallel in the parallel arrangement direction, and

among the standard lands arranged in parallel, the standard land arranged at least at one end in the parallel arrangement direction has a dimension that is increased in a direction away from another standard land adjacent thereto.

3. The transformer module according to claim 1, wherein each solder layer is joined to an entirety of the corresponding bent portion.

4. The transformer module according to claim 1, wherein the connection land connects the standard lands adjacent to each other in the parallel arrangement direction over an entirety of the standard lands in a direction orthogonal to the parallel arrangement direction.

* * * * *